

Recursion Diagnostic Test: Solutions

The results do not count for anything. This test is to allow the ECS401 team to diagnose if you need help to ensure they do not fall behind in learning to program. Understanding how for loops work is critical so if you have any problems we want to sort them out now. You must show all your working in the following. No correct table (exactly the same)... no marks.

1. What is printed after a method call `keepgoing2(5);`

```
public static int keepgoing2 (int n)          // L1
{
    if (n<=1)                                // L2
        return 3;                            // L3
    else
    {
        int answer = 5 + keepgoing2(n-2);    // L4
        System.out.println(answer);         // L5
        return answer;                       // L6
    }
}
```

Complete the dry run tables overleaf (you may not need all rows or all tables):

keepgoing2 (5)						
LINE	n	n<=1 ?	answer	keepgoing(n-2) =?	return	printed
L1	5					
L2		5 <=1 ? false				
L4			13	keepgoing(3) =? 8		8
L5						13
L6					13	

keepgoing2 (3)						
LINE	n	n<=1 ?	answer	keepgoing(n-2) =?	return	printed
L1	3					
L2		3 <=1 ? false				
L4			8	keepgoing(1) =? 3		
L5						8
L6					8	

keepgoing2 (1)						
LINE	n	n<=1 ?	answer	keepgoing(n-2) =?	return	printed
L1	1					
L2		1 <=1 ? true				
L3					3	

Write exactly what is printed when the method is called in the box.

8
13

2. What is printed after a method call `keepgoing2(7);`

```
public static int keepgoing2 (int n)          // L1
{
    if (n<=1)                                // L2
        return 3;                             // L3
    else
    {
        int answer = 5 + keepgoing2(n-2);    // L4
        System.out.println(answer);          // L5
        return answer;                        // L6
    }
}
```

Complete the dry run tables overleaf (you may not need all rows or all tables):

keepgoing2 (7)						
LINE	n	n<=1 ?	answer	keepgoing(n-2) =?	return	printed
L1	7					
L2		7 <=1 ? false				
L4			18	keepgoing(5) =? 13		8 13
L5						18
L6					18	

keepgoing2 (5)						
LINE	n	n<=1 ?	answer	keepgoing(n-2) =?	return	printed
L1	5					
L2		5 <=1 ? false				
L4			13	keepgoing(3) =? 8		8
L5						13
L6					13	

keepgoing2 (3)						
LINE	n	n<=1 ?	answer	keepgoing(n-2) =?	return	printed
L1	3					
L2		3 <=1 ? false				
L4			8	keepgoing(1) =? 3		
L5						8
L6					8	

keepgoing2 (1)						
LINE	n	n<=1 ?	answer	keepgoing(n-2) =?	return	printed
L1	1					
L2		1 <=1 ? true				
L3					3	

Write exactly what is printed when the method is called in the box.

8
13
18

3. What is printed after a method call `keepgoing2(0);`

```
public static int keepgoing2 (int n)          // L1
{
    if (n<=1)                                // L2
        return 3;                            // L3
    else
    {
        int answer = 5 + keepgoing2(n-2);    // L4
        System.out.println(answer);          // L5
        return answer;                       // L6
    }
}
```

Complete the dry run tables (you may not need all rows or all tables):

keepgoing2(0)						
LINE	n	n<=1 ?	answer	keepgoing(n-2) =?	return	printed
L1	0					
L2		0 <= 1 ? true				
L3					3	

LINE	n	n<=1 ?	answer	keepgoing(n-2) =?	return	printed

Write exactly what is printed when the method is called in the box.

No recursive calls are made and nothing is printed, just the value 3 is returned. Only one table is needed. Remember to fill out the call being dry run in the table with argument(s) in the header row of the table.